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CREATE

TECHNOLOGICAL PROTECTION MEASURES AND DIGITAL PRESERVATION

Evidence from Video Games

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About the Three Reports:

This report is the third in a three-part series assessing the legal status and providing evidence on Technological Protection Measures (TPMs). Report 1 by Anthony Rosborough (2024) provides a comparative doctrinal analysis of TPMs in the 27 Member States of the European Union plus key comparator jurisdictions including the UK. That document contains a detailed analysis of the status of TPMs and copyright exceptions, and different national approaches taken inside and outside of Europe to ensure access to works for beneficiaries of exceptions and limitations to copyright. Report 2 provides evidence on the effects of TPMs for researchers, libraries and archives, gathered from a survey of researchers and institutions. The results offer insight into specific impediments to research, education and preservation, as well as the outcomes of legal requests to rightsholders to provide access. Report 3 (this document) provides economic evidence of the costs of dealing with TPMs in the preservation of digital materials. Quantitative data is analysed from the MAME video game emulation project, revealing significant labour costs and missed use arising from the need to circumvent TPMs to preserve digital materials, an issue of relevance for a range of digital preservation and research contexts.

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EXECUTIVE SUMMARY

This report provides an empirical analysis of the impact of Technological Protection Measures (TPMs) on preservation of digital materials. TPMs impose costs from an educational, research and cultural heritage perspective. In this report, we assess costs represented by delays introduced from TPMs to the successful preservation of historical video games. Although the study is focused specifically on video game preservation, the approach and findings are relevant to understanding impacts of TPMs on all types of digital material, including images, audiovisual works and electronic records that research and cultural heritage institutions are required to preserve in order to fulfil their mission.

The study is focused on the Multiple Arcade Machine Emulator (MAME), the longest-running open-source emulation project, used by individuals, research and preservation institutions as well as game companies. The specific findings from this study are:

- ▶ Born-digital materials present specific challenges for preservation and research, particularly due to obsolescence of physical hardware that may include TPMs.
- ▶ TPMs can inhibit both format-shifting and emulation approaches to preservation of digital materials.
- ▶ The MAME emulation initiative has successfully emulated over 14,377 legacy electronic devices, including 3,783 arcade machines dating from 1979–2023.
- ▶ Of the 3,783 video games studied in our sample, 712 required circumvention of TPMs to to enable preservation.
- ▶ We measured the time delay introduced by the need to circumvent TPMs to the preservation effort, finding that this added an additional 10.6 months per item to preservation.
- ▶ Taking the 712 games from our sample that required circumvention, this represents 628 additional years required to preserve those games protected by TPMs. This represents a social cost in terms of additional labour from community volunteers, as well as missing use from the absence of preserved copies of games available for research, innovation and other productive uses. Moreover, as preservation is delayed technical, costs for knowledge institutions may rise.
- ▶ Digital materials that depend on TPM-protected formats and devices can become permanently inaccessible to research and preservation when rightsholders or TPM manufacturers can no longer be located.

To achieve our aims, we performed data mining of software development notes from the MAME project, augmented by metadata about original game releases. This approach enabled us to detect whether TPMs were present on original devices, and whether MAME volunteers undertook any circumvention of TPMs to successfully preserve individual games. Taking the span of time between original release and successful emulation in MAME, we statistically model the impact of the need to circumvent TPMs on the preservation effort.

We find that legacy TPMs present a considerable and statistically significant impediment to preservation. Controlling for factors such as original manufacturer, complexity of the underlying game and the presence of other challenges such as graphics and sound, we find that circumvention of TPMs added an additional 10.6 months (0.883 years) to the time required to preserve a work.

Taking the 712 games from our sample that required circumvention, this represents 628 years of social cost from TPMs. The full MAME catalogue of emulated devices is much larger (14,377 devices), meaning that total costs to society are even higher.

Social costs include not only labour by MAME contributors and researchers, but also losses from the unavailability of preserved games for research, education and innovation. As a comprehensive and interoperable archive of video game history, MAME is used by a variety of beneficiaries including traditional memory institutions, computer hardware researchers, enthusiast communities and follow-on commercial innovators. Delays to successful preservation produce negative effects for diverse user communities and society, not least for education, research and cultural heritage purposes (e.g. researchers wishing to study the history of video game programming).

TPMs inhibit the positive effects that would normally result from preservation, not least the benefits that come from works enriching the public domain in due course. For those games still under copyright, without an accessible working copy, there can be no productive exercise of exceptions, nor can there be research. Successful preservation benefits innovation, because of the role played by memory institutions and archives as resources for research and development. Many software companies have relied on emulated backups of their own copyright works when re-releasing them as new products, evidencing the benefit to innovation that is threatened if preservation is inhibited.

This report recommends that law and policy take account of future social costs of TPMs and weigh the potential benefits of statutory exceptions for circumvention to aid preservation, research and interoperability. Our specific policy recommendations are:

- 1/** to clarify that cultural heritage institutions may circumvent TPMs for preservation purposes without having to navigate the cumbersome administrative processes currently imposed by European copyright law;¹ and
- 2/** to weigh the future costs of TPMs against strong protections granted to rightsholders, for example by requiring that rightsholders maintain and offer robust means for European institutions and researchers to access TPM-protected works into the future.

Given the uptake of new business models and distribution methods such as cloud and software-as-a-service (SaaS), born-digital materials are at greater risk than ever from inaccessibility. TPMs will continue to inhibit preservation of digital materials well into the future, so policy action to mitigate their impacts and support European memory institutions is vital.

¹ See Report 1: Rosborough, A. (2024) Technological Protection Measures and the Law: Impacts on Research, Education and Preservation. Fig 11.

INTRODUCTION

Technological protection measures (TPMs) pose considerable challenges for education, preservation and research (Akester, 2009; McDonald et al., 2015; Erickson & Stobo, 2024; Rosborough, 2024). Due to legal uncertainty, exacerbated by lack of a harmonised and effective approach to exceptions in European law, institutions and researchers are often unsure about what can be done to gain access to TPM-protected materials when carrying out work that would normally benefit from an exception to copyright.

TPMs can impede the usability of materials, with negative consequences for research, innovation and economic activity. Ultimately, TPMs may prevent works from entering the public domain if prospective lawful users cannot effectively gain access to them. Preservation of “born-digital” materials may face the greatest challenge from TPMs, given their inherently fragile and unstable nature (Corujo et al., 2016; Klein & Whyte, 2022). Since digital media is often tied to a specific hardware or software environment at the point of creation, those materials become more difficult to preserve as formats obsolesce (Ippolito, 2016; Lemley, 2021; Farrand, 2022). TPMs intended to prevent unauthorised use at the time of initial publication can outlive the economic benefit they provided to rightsholders, while still causing difficulties for preservation. Digital materials that depend on TPM-protected formats and devices can become permanently inaccessible to research and preservation when rightsholders or TPM manufacturers can no longer be located.

This report addresses the lack of empirical evidence about the impact of TPMs on the use of digital materials by analysing original quantitative data about impediments to video game preservation represented by TPMs. The focus of this study is the Multiple Arcade Machine Emulator (MAME), the longest-running volunteer open-source video game emulation project. Although the report focuses on video games, the findings are relevant to understanding the impact of TPMs on a range of important digital materials, including images, audiovisual material and electronic records that may only be accessible via emulation.

Video games are something of a “canary in the coalmine” for digital preservation, signalling wider problems afflicting the preservation sector. Video game preservation has benefited from enthusiast communities like MAME that undertake significant labour and may operate outside of the law. Preservation of more mundane electronic records – located in cultural heritage institutions that are constrained by copyright law – is much more challenging. Without timely intervention, facilitated by an appropriately flexible copyright regime, costs for knowledge institutions are increased and we prejudice not only education and research but the ability of future generations to access their own digital cultural heritage.

During their preservation work, MAME volunteers have encountered hundreds of different TPM protections on legacy video game devices that required circumvention before preservation could be completed. The MAME initiative was chosen as a study site because the open-source code base and public documentation around MAME permits quantitative study of the challenges facing video game preservation. Our methodological approach is intended to inform broader studies and policy considerations about the impact of TPMs on a diverse range of digital materials, not only video games.

The emulation approach adopted by MAME is already being applied in a range of preservation and research contexts for which the impact of TPMs is relevant. For example, museums have used emulation to re-create lost exhibitions of digital art originally made on 1980s microcomputers.² Researchers and historians have used emulators to study the history of the electronics and interactive entertainment industry,³ while video game publishers⁴ have even used emulation to access their own original software lost to the passage of time (Ippolito, 2016; Ponsard, 2021).

Emulators like MAME allow users to run computer programs on hardware that they were not originally intended for, approximating the original software experience as closely as possible in a new user environment. Emulation therefore opens use of a wide range of digitised and born-digital materials, including computer programs, electronic documents, interactive multimedia and digital image files, among others. Emulation carries copyright risk, since emulators may be used to run unauthorised copies of copyright-protected material, and because they may require access to protected device firmware in order to function. However, as an informal volunteer organisation, MAME has not been encumbered by legal and reputational risks facing traditional institutions. The project has managed to operate without legal protections such as DMCA anti-circumvention exemptions available to formal cultural heritage institutions in the USA. For the most part the project's activities have been tolerated by rightsholders.

MAME occupies an unusual position between traditional preservation institutions and the public, serving a variety of users including video game enthusiasts, technologists, historians, cultural heritage institutions and follow-on innovator firms. One possible reason that rightsholders tolerate MAME is the service it provides by maintaining access to games and making them interoperable on modern devices, a valuable resource for rightsholders seeking to re-release old games from their catalogues. Another reason is that many of the devices emulated in MAME date from the coin-operated arcade era (1978–2000), with many of those firms now defunct or rightsholders no longer interested in pursuing enforcement. One recent study found that 87% of video games released prior to 2010 were no longer commercially available (Salvador, 2023).

As a preservation effort, MAME faces a number of challenges. The project has limited resources, mainly relying on volunteer efforts of skilled programmers to complete its work. The project has also had to locate working examples of hardware in order to reverse-engineer and preserve them, sometimes at significant cost. MAME developers have had to contend with technological protection measures consisting of purposeful techniques used by manufacturers to prevent access to the software code contained in memory on printed circuit boards (PCBs). Progressively more elaborate TPMs included encryption via memory controller units (MCUs), encrypted ROMs with authorisation codes unique to every board, and so-called "suicide" chips that would wipe their contents if tampered with.

² For example, the Guggenheim Museum investigated a series of case studies to formulate creative strategies for preserving endangered works. One work chosen to test emulation was Grahame Weinbren and Roberta Friedman's video installation *The Erl King* (1982-85), which used "a combination of obsolete hardware, artist-written software and custom-made components." (Jones, 2004).

³ For example, researchers Helen Stuckey and Melanie Swalwell (2014) describe how they used fan emulation sites to gather information about text adventure game *The Hobbit* (1982) by Beam Software.

⁴ *Capcom Arcade Stadium*, a retro game collection released in 2021 for Nintendo Switch, uses MAME to emulate games from the publisher's past catalogue.
Source: <https://datahorde.org/made-with-mame-capcom-arcade-stadium/>

During the profitable and competitive arcade era, these measures were considered necessary by some manufacturers to control the distribution of their software and prevent unauthorised bootlegs from rival firms. However, in the present era these legacy TPMs pose a major challenge to preservationists seeking to document and preserve video game history.

The purpose of this study is to investigate the social costs represented by legacy TPMs on old arcade video game hardware. We hope that increased understanding of the cost of TPMs for future preservation can inform debates about wider challenges for digital preservation in regards to all types of content that are important for cultural heritage, education and research. Many of the challenges identified in this study are relevant for researchers and historians working in neighbouring domains, such as preservation of digital art or archiving of electronic documents which depend on TPM-protected hardware to be accessed. Technological protections of today will become the legacy TPMs of tomorrow, with downstream social costs that are not fully understood or accounted for in policy discussions.

The report proceeds as follows:

- Section 2** offers a review of the literature on digital preservation and the social costs introduced by TPMs.
- Section 3** describes the MAME project in detail and highlights some of the specific issues arising from TPMs, to identify the technical challenges facing preservation.
- Section 4** discusses the research method, which consisted of quantitative analysis using computer-assisted data mining of publicly available data about the contents and development of MAME.
- Section 5** presents the results of the quantitative data collection and regression analysis, which investigate the effects of legacy TPMs to identify delays in successful preservation introduced by the presence of TPMs.
- Section 6** offers policy recommendations;
- Section 7** discusses limitations of the study and
- Section 8** concludes with insights for future research.



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2. LITERATURE REVIEW

In addition to the challenges already involved in preservation of analogue media, preservation of “born-digital” objects involves new complexities and costs (McDonough et al., 2010; Farrand, 2022; Klein & Whyte, 2022). From an institutional and organisational perspective, digital preservation raises new management concerns and resource requirements. The Open Archival Information Standard (OAIS, 2012) sets out the parameters for sustainable governance of digital archives from the point of ingest, to management of the information (including archival storage and administration) through to decisions about access standards for dissemination to users. It is therefore a useful framework for assessing the costs related to preservation of digital objects over their full lifecycle.

In a meta-analysis of archival practices, Corujo et al. (2016) identify a number of specific costs related to born digital objects used in research, including: (1) appraisal and selection of materials for preservation; (2) quality control and maintenance of stored materials; (3) storage itself including technological formats, hardware, energy costs, and institutional overheads; (4) access, use and re-use within the organisation and by external stakeholders. Successful preservation of digital objects often requires specialist skills and training as well as labour costs related to processing and contextualisation of digital information for effective re-use (Bachell and Barr, 2014; Corujo et al., 2016). Many born-digital collections such as video games lack basic metadata standards to describe and manage them and those need to be developed for each new type of media preserved (McDonough et al., 2010).

Even without considering the possible impediments introduced by copyright TPMs, digital preservation is already technologically complex and costly. Evaluating impediments to digital preservation of historical films, Antoniazzi (2021) found significant costs for institutions related to the need to migrate film contents away from obsolete formats, exacerbated by industry practices of planned obsolescence and technological change. Vulnerability emerges from reliance on commercial solutions (such as film scanners) which can disappear from the market, leaving institutions in an unsupported position. Notable additional costs for digital preservation include skilled labour costs, high infrastructural maintenance costs related to frequent migration and updates, and higher electricity costs from data storage. While storage costs have decreased in a manner approximating Moore’s Law,⁵ they remain significant and have not decreased below that of preserving analogue films. Digital film preservation was estimated to be more expensive than analogue methods: The long-term cost of digital preservation was estimated to be up to 11 times more expensive than analogue film preservation, with the Academy of Motion Picture Arts and Sciences (2007) estimating the cost of preserving a 4K resolution digital feature film at \$12,514 per year, including storage and maintenance (Antoniazzi 2021).

⁵. See ‘What is Moore’s Law and is it still relevant today?’ BBC Science Focus Magazine: <https://www.sciencefocus.com/future-technology/moores-law>

Video game preservation introduces additional costs related to diverse storage media and technologies. Many video games straddle the divide between born-digital and analogue preservation objects, comprising both a physical medium (such as a cartridge, circuit board, magnetic or optical storage disk) as well as a digital object (the software code and game assets) separate from the physical layer. Increasingly, video games are fully digital objects, distributed via online storefronts and mobile app stores, and played over online servers (Lee, 2018; Lemley, 2021; Salvador, 2023). Preservation faces the problem of obsolescence of the underlying storage media as well as the software and hardware environments needed to run games (McDonough et al., 2010; Farrand, 2022). There are generally two approaches to preservation of video games: migration and emulation (Ponsard, 2021). The former approach seeks to migrate the data from underlying physical media to new formats in order to avoid deterioration ("bit rot"). This method is commonly used in film preservation, for example when migrating old film reels at risk of chemical deterioration to new formats such as linear tape-open (LTO) (Antoniazzi, 2021). In video game preservation, Lowood et al. (2009) estimated that the games most susceptible to deterioration were those stored on magnetic disks from 1985 and earlier. The second approach and the one taken by the MAME community is to virtually reproduce or emulate the original hardware environment needed to run the program data, setting games free from specific storage media. While emulation has not been widely used by institutions, it may be the most effective long-term solution because of the cost and difficulty of maintaining collections of physical hardware and storage media (Ippolito, 2016). However, both migration and emulation strategies can be impeded by copyright law and technological protection measures (Farrand, 2022; Cross, 2023).

2.1 IMPEDIMENTS TO DIGITAL PRESERVATION ARISING FROM COPYRIGHT

In addition to the costs identified above, copyright law introduces additional impediments to successful preservation of digital objects. Since most born-digital objects were created in the previous 70 years, the vast majority are still in copyright and will remain so for a significant length of time (Farrand, 2022). Many preservation activities will therefore run into challenges from copyright law, depending on what is required to successfully preserve a digital object. Preservation can involve a range of activities, including straightforward collection of artifacts (museum approach), migration, emulation, re-creation of lost works using new technologies, and preservation of ephemera such as development notes and fan works, all of which introduce complexities from a copyright perspective (Lowood, 2009, Stuckey and Swalwell, 2014). Taking two long-term digital preservation approaches, migration and emulation, we find specific areas in which copyright law and anti-circumvention provisions may inhibit preservation of video games.

Because migration requires "wholesale copying" from one storage medium to another, the process is likely to infringe copyright, absent an exception covering the preservation activity (Lee, 2018). Successful migration may also require circumvention of TPMs that control access to or copying of game code. This can prevent even basic preservation if institutions are not legally or technically able to make preservation copies, access works to make use of them for research purposes, or undertake migration.

Emulation is even more complex from a copyright perspective, because it requires reverse-engineering of both the games and the hardware environments needed for them to function. Emulation may require decompilation and examination of parts of a system's firmware. The emulation process involves "multiple acts of reproduction and adaptation, all of which are potentially infringing" (Lee, 2018). Reverse engineering of software is permitted under Article 6(1) of the Software Directive, which permits decompilation of software "to obtain the information necessary to achieve the interoperability of an independently created computer program with other programs". However, as pointed out by Lee (2018), the creation and distribution of an emulator may give rise to accessory liability for infringements carried out by users of the emulation tools (2018).

In Europe, the Information Society (InfoSoc) Directive obliges Member States to require rightsholders to take voluntary measures to enable beneficiaries to benefit from a closed list of exceptions, including under Article 6 CDSM Directive a mandatory exception for purposes of preservation, although the means of benefiting from exceptions varies by country. Member States may intervene by taking "appropriate measures" when rightsholders fail to provide effective means of benefiting from privileged exceptions. The requirements outlined in the InfoSoc Directive apply only to specific exceptions, limiting the ability of the general public to benefit from other copyright exceptions in their specific jurisdiction and creating a confusing patchwork of legal uncertainty across Europe (Renda et al., 2015; Rosborough, 2024). Furthermore, the requirement in EU law for prospective beneficiaries of exceptions to seek access from rightsholders raises an "orphan TPMs" problem. If rightsholders cannot be located, or if third-party TPM providers are no longer in business, it may be impossible to gain access without circumvention.

In the United States, new exemptions to the anti-circumvention provisions contained in DMCA §1201 are considered on a triennial basis. While previous rulemaking has created opportunities for specific classes of users (such as cultural heritage institutions) to circumvent TPMs to make specific uses of works (for example preservation copies), the approach has the weakness that rules are only valid for the three-year period following their adoption and must be renewed by the Librarian of Congress, resulting in an unstable legal environment for institutions concerned with long-term preservation strategies (McDonough et al., 2010). The approach taken by the USA also does not address the technical challenge of circumvention, which we analyse in this report.

In 2015 the Library of Congress issued an exception to the DMCA §1201 for circumvention of TPMs to enable functioning of games relying on server authentication to function after servers are discontinued. The exemption was updated and broadened in 2018 and 2021. The rule enables eligible libraries, archives, and museums to circumvent technological protection measures on lawfully acquired computer programs (including video games) when they are no longer reasonably available in the marketplace, for preservation purposes. Under the triennial rulemaking of 2021–2024, institutions may additionally make single copies available off-premises to a single user at a time. The overall direction of travel in anti-circumvention rulemaking in the USA suggests broadening recognition of the importance of digital preservation, the impediments to lawful use presented by TPMs, and the specific requirements of preservation institutions, although significant opposition from commercial rightsholders remains (Albert, 2018; Sigmon, 2023).

2.2 SOCIAL COST OF TPMs

Preservation is one activity that can be impeded by the presence of TPMs, but there are potentially wider social costs to consider. Rothchild (2006) argues that the application of TPMs imposes undue costs (externalities) on users who were not party to the initial transactions that gave rise to the costs (e.g. the initial sale of the good carrying TPM protection). This promotes an oversupply of TPMs, since manufacturers do not bear the full cost of these externalities, but reap the benefits of stronger protection at the point of initial release of their products (2006).

According to Rothchild, TPMs produce various negative externalities: (1) they lead to a contraction of the public domain, including the ability for lawful users to enjoy certain exceptions, as well as access parts of works that would otherwise be available to researchers (e.g. portions of a work on the idea side of the idea/expression dichotomy, public domain content etc.). In the case of electronics research such as that carried out by MAME, TPMs can inhibit access to information that enables interoperable uses of the underlying work. The long-term impact of TPMs may even exceed the initial copyright term of the work, as will be the case where works cannot be circumvented and therefore become entirely inaccessible. (2) TPMs may also cause reduction in the market of used copies of works. Specifically, TPMs that tether a work to a specific user or device are detrimental to the secondary market by preventing transfer of the good. This is the case for certain video game TPMs, for example those that link a copy of a game to a specific user or device. (3) TPMs can potentially interfere with competition in the market. For example, device manufacturers can use TPMs to prevent competitors from making interoperable products, locking them out of the market. In the field of video games, TPMs also inhibit reverse-engineering and research that could lead to complementary products or services. To the categories of negative externality proposed by Rothchild (2006), we propose a fourth: (4) TPMs inhibit the positive spillovers that would normally result from preservation. Successful preservation is linked to the other externalities and is a prerequisite for works to be accessible in the public domain. Without an accessible working copy, there can be no productive fair dealings, nor can there be research. Preservation also impacts innovation, because of the role played by memory institutions and archives as resources for research and development. The fact that many software companies have relied on emulated backups of their own copyright works when re-releasing them as new products, is evidence of the benefit to innovation that is threatened if preservation is inhibited.⁶

A full accounting of costs and benefits from TPMs is required for policy to make informed choices and for institutions to plan for future requirements. Corujo et al. (2016) argue that the lack of empirical data about costs of digital preservation limits predictive capacity and therefore the ability to manage preservation efforts. Budgeting for all costs (including those from TPMs) is required for memory and research institutions to make the most efficient use of limited resources and plan future investment. Table 1 summarises the main costs and impediments to preservation of born-digital objects arising from TPMs, for both organisations and user communities (researchers, innovators and users).

Impediments and costs from TPMs for digital preservation		
Community	Source of impediment/cost	Main studies
INSTITUTIONS	► Ingest: obtaining usable accessible materials otherwise restricted by TPMs	McDonough et al., 2010; Corujo et al., 2016
	► Preservation of materials, including making multiple copies for redundancy	Lowood et al., 2009; Klein & Whyte, 2022
	► Maintaining access to materials connected to obsolete formats and devices	McDonough et al., 2010; Lee, 2018; Klein & Whyte, 2022
	► Research uses including text-and-data mining, manipulation and analysis	Corujo et al., 2016
	► Enabling public use of TPM-protected materials (e.g. limits on number of copies accessed or used)	Corujo et al., 2016; Klein & Whyte, 2022
	► Human resource costs of maintaining collections (expertise, training)	Butler et al., 2019; Antoniazzi, 2021
	► Costs arising from legal uncertainty about copyright exceptions and costs of establishing legal compliance for new activities	Renda et al., 2015; Lee, 2018; Butler et al., 2019; Cross, 2023
USER COMMUNITIES	► Impediments to preservation of personal data and user-generated digital content	Lowood et al., 2009; McDonough et al., 2010
	► Impediments to innovation, interoperability and complementary product development	Rothchild, 2006; Lemley, 2021
	► Impediments to freedom of expression, research and private study	Rothchild, 2006

Table 1: Impediments and costs from TPMs for digital preservation



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3. MAME AND VIDEO GAME PRESERVATION

The MAME project has been active since 1997 and has benefited from the contributions of hundreds of volunteers from around the world. The project now comprises over 10 million lines of code, enabling support for over 14,700 unique hardware devices. The project's website describes its mission as follows: "MAME's main purpose is to be a reference to the inner workings of the emulated machines. This is done both for educational purposes and for preservation purposes, in order to prevent historical software from disappearing forever once the hardware it runs on stops working."⁷

Considering its cultural and technological significance, there is surprisingly little academic research about MAME. Some notable exceptions include Murphy (2013) who discusses the ways in which MAME disrupts traditional archives while serving an important decentralised archival function, and Schiavone et al. (2015) who discuss how MAME helps to bridge between familiar older technologies (vintage games) and newer technological formats that support emulation. The purpose of the present report is to provide empirical insight into the labour involved in the MAME project, specifically the effort and resources required to successfully preserve games protected by TPMs.

MAME is an open-source software project, but its participants also view their work as a form of preservation (Giles, 2008). As noted by Murphy (2013) MAME's approach to preservation is unconventional and focused on use rather than static archiving. It also serves an enthusiast community by enabling gamers to play arcade machines and other games on their home devices. The expanding catalogue of emulated devices was therefore driven by the intellectual pursuit of knowledge of old hardware as well as a desire to preserve fondly remembered game machines. There is no systematic sampling method to the prioritisation of devices awaiting emulation (for example, the project could have started from *Spacewar!* (1962) and worked forwards in time). Rather, the selection process was influenced by both popularity of games and ease of emulation. Due to complexity of later game hardware and the capabilities of home computer at the time, the early compatibility list shows a bias towards older and less complex titles.

⁷ About MAME. Accessed online: <https://www.mamedev.org/about.html>

MAME was started by solo programmer Nicola Salmoria, whose first emulator project, MultiPac, supported a collection of games that all used similar *Pac-Man* (1980) hardware. MAME version 0.1, launched in February 1997 initially supported 5 games based on *Pac-Man's* Zilog Z80 architecture. Prior to the initial release of MAME, emulators existed for some 8-bit home consoles and single games like *Donkey Kong* (1981) and *Space Invaders* (1978) but many coin-operated arcade games used sophisticated hardware that was harder to emulate. The early focus of MAME was to support popular coin-operated arcade titles and those that used similar standardised chip architecture. As new chipsets became possible to emulate, batches of new game machines could be supported by the project. Compatibility expanded as improvements in home PCs enabled emulation of more complex machines. Capcom System 1 (CPS1) support, based on the Motorola 68000 was added in 1998. Support for 3D hardware such as Namco's System 22 came later in 2005. MAME version 0.258, released on 30 August 2023, now provides emulation support for some 14,377 unique devices, including arcade machines (the initial focus), home computers, home consoles, gambling slot machines, bar-top quiz devices, handheld toys, music synthesisers and assorted peripheral devices. Figure 1 shows current compatibility for 3,945 coin-operated arcade games currently emulated in MAME.⁸

Very early in the project, MAME programmers began encountering and taking note of copyright protection measures on arcade circuit boards. Figure 1.1 shows the prevalence of TPMs detected by the MAME development team in game releases over time. Sometimes the developers referenced specific encrypted microprocessors, such as the Hitachi FD1094, while in other cases they noted only the general presence of copyright protection or security measures. As MAME development continued, dealing with TPMs would occupy a significant amount of programmer attention, often eclipsing challenges posed by emulation of graphics, sound or gameplay. In the 1990s, one of the earliest TPMs that posed challenges for preservationists was the "Slapstic" protection chip included on many 16-bit Atari games such as *Marble Madness* (1984), *Gauntlet* (1985) and *Paperboy* (1985). MAME developer Aaron Giles noted that the chip was unique to every Atari game and used to prevent operators from burning new EPROMs and "upgrading" their PCBs to a new game without buying official upgrade kits from Atari (Giles, n.d.). In 1998, Giles successfully reverse engineered functionality of the Slapstic security, enabling full emulation of the 8- and 16-bit Atari arcade games that used the protection. Protection measures became increasingly sophisticated. The Hitachi FD1089 / FD1094 found on Sega System 16 boards used an encryption key to verify the contents of a game before running the code. The chip required battery power to operate correctly, so when batteries died the chip would no longer function, resulting in a fault. Because such systems rely on inaccessible sources of power inside TPM modules in order to function, they have been dubbed "suicide boards" by the preservation community. In 2018, researcher Eduardo Cruz published a method for backing up and restoring ("de-suiciding") Hitachi FD1094 modules. He indicates that circumvention required work by several individuals over a span of years:

— **This guide is the result of several years of work by a small group of arcade enthusiasts to unravel the secrets of the security implementation found in one of the most loved and popular arcade game platforms. Thanks to this work it is now possible to fully preserve most Sega 16-bit systems enabled with security as fully working unmodified originals (Cruz, 2018).**

⁸. All data from this study, including the presence of TPMs and emulation support for 14,377 devices in MAME v0.258 can be accessed using our public data visualisation tool: https://copyrightevidence.org/2024/MAME_TPMs_Explorer/

While Sega System 16 games that used the FD1098 / FD1094 CPU were functional in MAME prior to 2018, this was made possible without full emulation of the security measure. Such “hacks” might involve bypassing the security or otherwise making the game system behave as if the TPM was present. MAME developer David Haywood expressed that because the purpose of the MAME emulation project is to preserve and document the functioning of all game-related hardware, hacks that do not properly preserve the functioning of TPMs are the least-preferred option. Games with imperfectly emulated TPMs might exhibit non-standard behaviour or crashes. As of 2023, there were 162 devices in MAME with imperfect protection emulation (causing problems with graphics, sound or game logic). An unknown number of devices are unsupported because TPMs make emulation impossible.

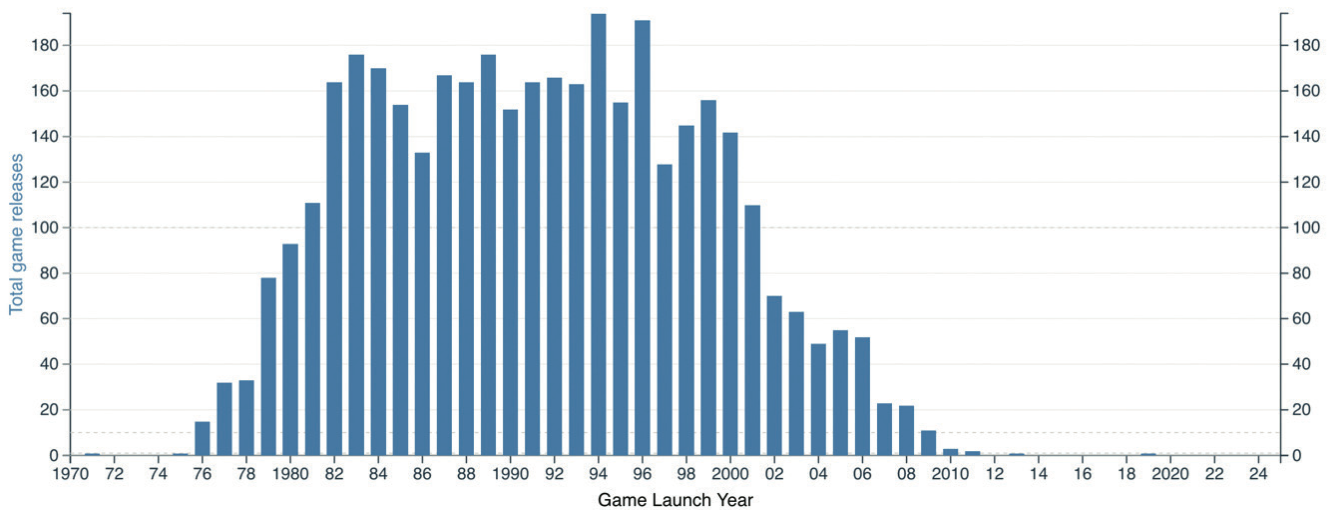


Figure 1: Arcade game machines emulated in MAME by year of original release

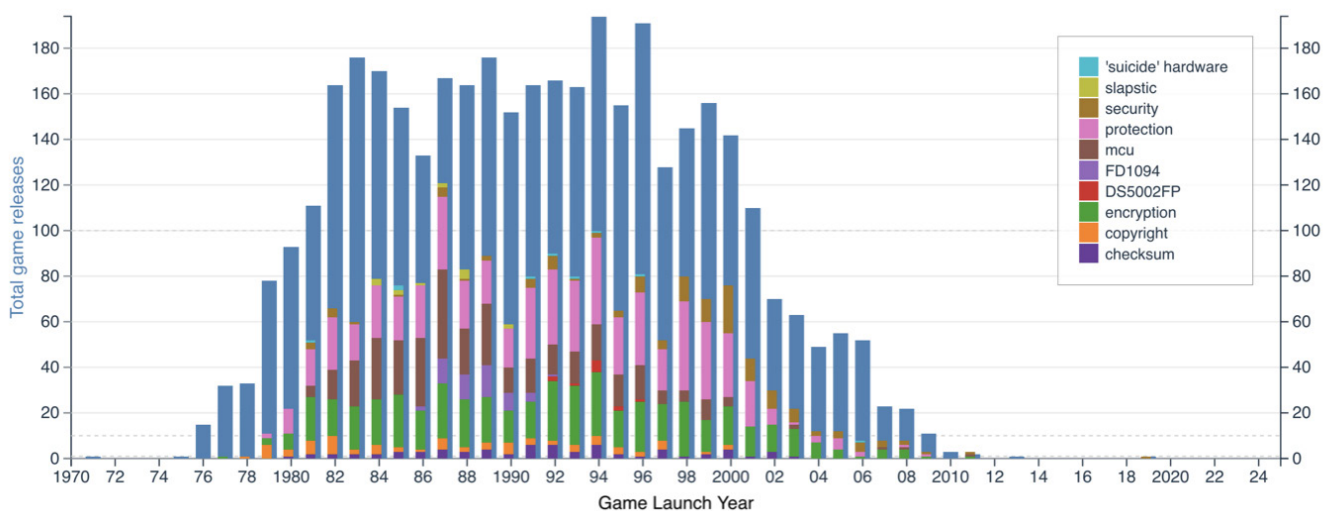


Fig 1.1: Types of protection encountered on devices⁹

⁹ For an explanation of technical terms, please see Glossary in Appendix 1

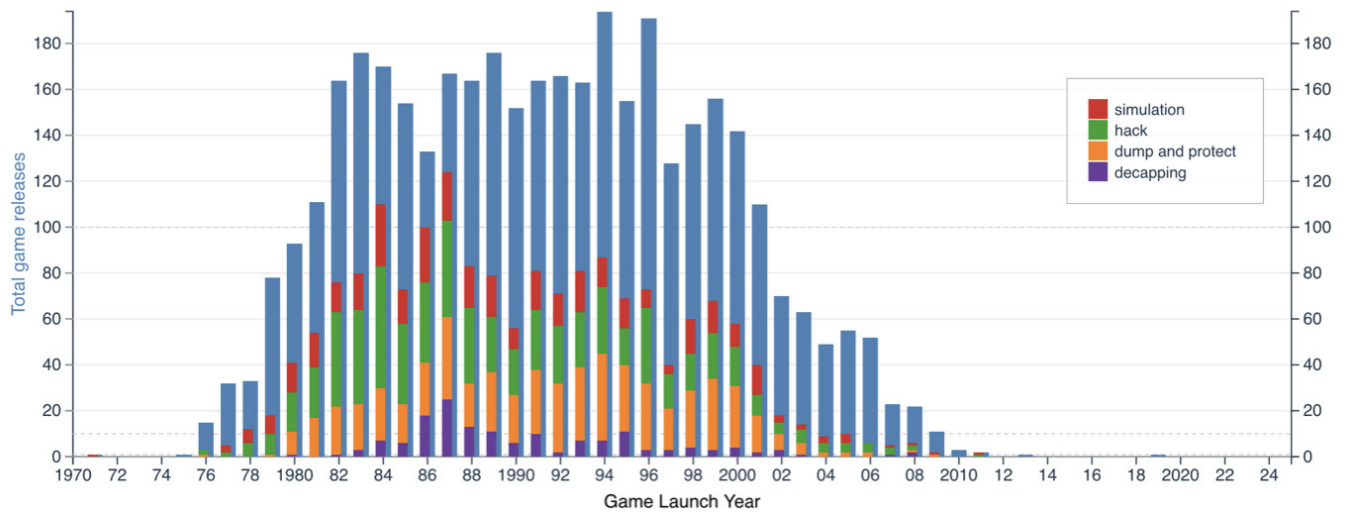


Fig 1.2: TPM recovery methods used

The addition in MAME of emulation support for Spanish arcade manufacturer Gaelco is illustrative of the challenges faced by MAME programmers from TPMs, and the potential for copyright locks to lead to total loss of digital heritage objects. Gaelco, which stood for *Gabinete Electrónico Consultivo*, was founded in 1984 in Barcelona Spain and achieved minor prominence as a manufacturer of coin-operated arcade games in Europe, releasing two dozen arcade titles between 1991 and 2005. Some of their games were hits, but many were less popular with few physical examples produced. Making matters harder for preservationists, Gaelco employed an unusual copyright protection measure on many of their circuit boards, the DS5002FP or “DALLAS” semiconductor. This TPM had sophisticated anti-tamper methods with code stored encrypted in a battery-backed SRAM chip. Because of the complexity and rarity of physical examples of the TPM, it took MAME volunteers and engineers until 2017 to successfully access and dump the contents of the DALLAS chip, finally enabling full emulation of most of the Gaelco game catalogue. Figure 2 shows the process used by the researchers to access and preserve the code from the TPM module. MAME developers describe their efforts as a “race against the clock” combatting the deterioration of arcade hardware.¹⁰ Some rarer games by Gaelco and other manufacturers may never be fully preserved due to TPMs.¹¹ In some cases the reverse-engineering process requires access to two original boards for comparison of encrypted contents. MAME developer Ryan Holtz explained to the authors that without discovery of a second arcade board, preservation of the Gaelco machine *Play 2000* will remain incomplete, since there is no way to be sure whether the single dump of code from the DALLAS chip is corrupted or missing data, which could lead to imperfect emulation behaviour.

¹⁰. Giles, A (2008).

¹¹. Another game nearly lost due to TPM protection is Last Survivor by SEGA. In 2012 a physical circuit board came up for auction in Japan. The board used the FD1094 suicide protection measure. Fortunately, due to MAME programmers’ experience with that TPM, the game was successfully preserved.
Source: <https://aarongiles.com/old/?p=241>

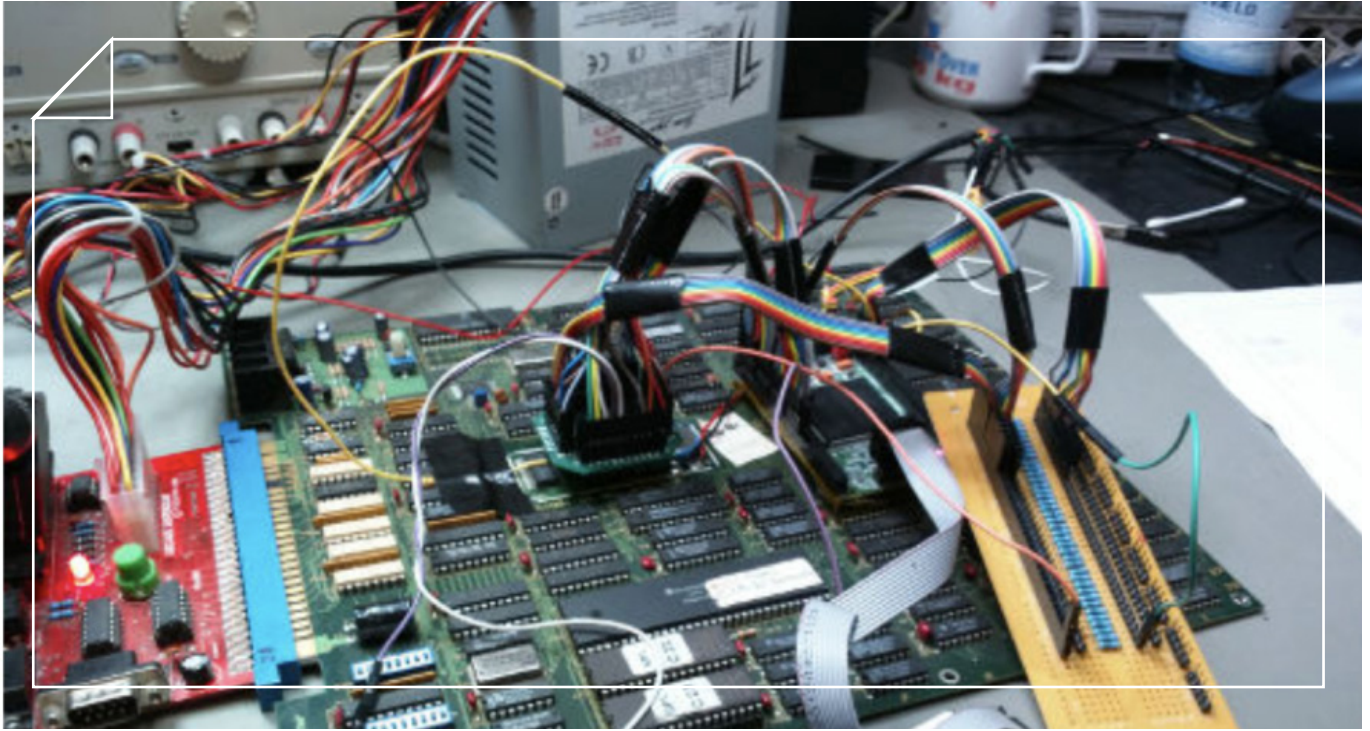


Figure 2:
Reverse engineering by Wilhelmsen and Kirkegaard (2017) to access the "Dallas" DS5002FP semiconductor and preserve operation of the module.



Steve Dimatteo / Unsplash

4. RESEARCH METHODS

In this study, we are interested in measuring social costs arising from the presence of TPMs in video games (hardware and software). To achieve this, we performed text and data mining on the MAME development repository and other open databases to gain insight into the emulation process and the labour involved in circumventing TPMs. The purpose of our study was to 1) detect the presence of TPMs on specific video game devices and 2) ascertain whether the need to circumvent TPMs introduced any delays in the preservation process, and to characterise those delays. One aspect of MAME that we exploit was the practice by developers of keeping highly detailed documentation of emulation progress. Maintaining rigorous documentation has enabled MAME developers to accumulate knowledge about preservation of machines and serves as a historical archive of computer architecture. Device-specific development notes included with the MAME software proved to be a valuable source of detail about TPMs encountered, as well as other technical challenges involved in the preservation process. In the following section, we describe in detail the data collection process and the variables used.

4.1 DATA COLLECTION

To gain a full picture of the MAME preservation project and successfully emulated devices, we collected data from several sources. First, we obtained a list of all currently supported devices in MAME by consulting the most recent ROMs set available on the Internet Archive.¹² At the time of the research, this was version 0.258 of MAME, released on 30 August 2023. The list was used to obtain the short game name (unique identifier) for every supported device, as well as the file size of each device ROM package. There were 14,377 individual devices supported in MAME version 0.258. This total includes arcade games (the main original focus of the project) as well as home consoles, personal computers, peripheral devices, gambling devices, quiz game machines, handheld devices and music synthesizers.

Working from the initial emulation compatibility list, we next added device information obtained from the volunteer online arcadeitalia.net database. This resource collects metadata from various sources, including game screenshots, control input, genre, manufacturer, date of original release and date of first inclusion in MAME. The Arcade Italia resource also helpfully parses the developer notes for each supported device contained in *mameinfo.dat*, a file that accompanies the MAME software and is updated with every new release. We used a cloud service and performed an automated web scrape on the full dataset, importing target variables to persisted datastore using NeDB.¹³ Data collection was spread over a two-week period to avoid overloading the server of the data hosting website. All device information was stored in a JSON file format with one device per line. We then used additional search of the database to identify presence of TPMs and constructed further variables using text mining of TPM-related keywords (see Table 2 below). The final dataset was compiled from three main sources: the MAME 0.258 ROM list obtained from Internet Archive; metadata on games obtained from Arcade Italia, and the parsed developer notes from *mameinfo.dat*, reproduced in the “Info” tab in Arcade Italia database.

¹² Latest ROM list was obtained from the Internet Archive: <https://archive.org/details/mame-merged>

¹³ <https://github.com/seald/nedb>

4.2 SAMPLE SELECTION

The complete MAME 0.258 ROM list comprises parent ROMs for 14,377 supported hardware devices, including arcade game machines, home consoles, personal computers, peripheral devices, handheld game devices, digital music synthesisers and other electronic devices using integrated circuit logic. "Parent" ROMs refer to a single definitive "dump" of device data, which may have alternate versions (such as prototypes or regional variations), referred to as "clones". To avoid duplication, we do not include clone ROMs in this study. Not all of the devices supported in MAME are fully emulated, with some still reporting imperfect sound, graphics, or copyright protection emulation, producing various bugs and glitches. The full list also includes a significant quantity of gambling devices (slot machines) made by various manufacturers. To focus the study and statistically isolate the effects of TPMs for preservation of a specific media type, we opted to sample only coin-operated arcade game machines from the full MAME supported device list. We made this choice on the basis that we are primarily interested in copyright-protected video games, rather than other materials such as PC device drivers or music synthesizer firmware. We avoided sampling home-console video game hardware because those devices were oriented towards a different market, could play large game libraries on separate storage media and were not the original focus of MAME emulation. Arcade video games were the primary focus of MAME development and are the products for which we observe the most intense application of TPMs by manufacturers (see Table 4).

Our sampling approach initially yielded 3,995 coin-operated arcade games spanning from 1971 to 2019. Many of the earliest arcade machines utilised transistor-to-transistor logic (TTL) architectures and did not implement copy protection systems. The earliest arcade machine to incorporate copy protection in our dataset was *Astro Fighter* by Data East, released in October 1979. Because this study is concerned with the presence of TPMs and their effects, we selected all games released from 1 January 1979¹⁴ to 30 August 2023 (the date of MAME version 0.258 release). This yielded 3,846 total game machines. Of those, a further 63 games had missing information about the year of original release. As we were unable to calculate our main dependent variable from those missing data, we excluded those results from our analysis, resulting in a final sample of 3,783 arcade game machines.

4.3 VARIABLES

We follow and Økland and Olsson (2020) and D'Alpaos et al. (2014) in treating units of time delay as the dependent variable to assess the impacts of different factors on any delays encountered. We compute a variable, *YEARS_DIFF* to capture the amount of time between a game's original year of release (ranging from 1979 to 2023) and its later support in MAME (ranging from 1997 to 2023). The range of this variable is from 0 (indicating that a game was emulated in the same year as its release) to 44 (the maximum amount of time permitted in our sample for a game initially released in 1979). *YEARS_DIFF* is the main dependent variable used in our regression analysis.

¹⁴. MAME developers noted that protection technique involved the EPROM datalines being routed through an 8bit x 256byte encrypted PROM. (MAME development notes, n.d.).

Independent variables are as follows. TPM_RECOVERY is the main independent variable of interest. It measures whether the MAME developers performed any circumvention of TPMs in order to recover device data and successfully preserve a game. Through discussion with MAME developers, we ascertained that there were four main methods of circumventing TPMs on arcade hardware: 1) directly reading or “dumping” the software contents of a security chip to enable its emulation; 2) simulating the presence of a TPM module to trick game software into performing normally; 3) otherwise hacking the device to bypass any TPMs on the original hardware; 4) physically removing or “decapping” the microprocessor or TPM module to obtain direct access to the machine code. We computed TPM_RECOVERY as a binary variable having a value of 1 if any of the following combination of keywords were detected in the developer notes for each game device: decap*+protect*, hack*+protect*, hack*+security, simulat*+protect*, simulat*+security, dump*+protect*, dump*+security. Wildcard keywords were used to capture permutations of the same word, e.g. “protected / protection”. It is likely that the total number of games requiring circumvention of TPMs is higher, but we opted to apply the strictest selection criteria to minimise false positives.

Variables used in study	
Variable name:	Definition
YEARS_DIFF	▶ Number of years span between a game’s original release and addition to MAME.
TPM_RECOVERY	▶ Binary variable indicating whether MAME developers undertook circumvention of TPMs.
GAME_AGE	▶ Age of game taken by subtracting its original release year from current year (2024).
MAME_AGE	▶ Age of game in MAME taken by subtracting the year of its first emulation from current year (2024).
ROM_SIZE	▶ Game file size in bytes. Used to control for game complexity.
PROBLEM_GRAPH	▶ Binary variable indicating whether there remain problems in emulating the game related to graphics.
PROBLEM_SOUND	▶ Binary variable indicating whether there remain problems in emulating the game related to sound.
PROBLEM_PROTCT	▶ Binary variable indicating whether there remain problems in emulating the game related to TPMs.
MAJOR_MFG	▶ Dummy variable indicating whether game was released by one of the top-12 manufacturers: Sega, Capcom, SNK, Konami, Nintendo, Taito, Atari, Namco, Midway, Williams, Centuri, Data East.
GENRE_SHOOT	▶ Dummy variable indicating the genre of game was “shooter”. Other categories were: ball & paddle, climbing, driving, fighting, maze, music, platformer, and sports.

Table 2: Variables used in study

We collected two key variables related to age of games and the date of earliest support in MAME. These are recorded as *GAME_AGE* (the year of original release) and *MAME_AGE* (year of first addition to MAME as a “working” ROM). These two variables were used to compute *YEARS_DIFF* above and are not included in the regression analysis to avoid multicollinearity. We also employ a variety of controls. The variable *ROM_SIZE* contains the size of individual ROM files in bytes. This variable serves as a control for overall complexity of games awaiting emulation. Larger games are presumed to introduce more complexity for developers to deal with, independent of any TPMs that may also be present on the device. An additional control variable, *LOG_GFLOPS* records the amount of compute per USD that could be bought by consumers in each year that MAME was active. This is to control for increasing speed of personal computers enabling emulation of more complex games and circumvention of TPMs with more powerful modern hardware. The variables *PROBLEM_GRAPH* and *PROBLEM_SOUND* record whether there are still any existing problems with the emulation of games, related to sound, or graphics. These are used as controls for presence of persistent challenging problems with boards, independent of any TPMs that may be present. This information was obtained from metadata parsed by the Arcade Italia database and captures whether aspects of a game are emulated imperfectly based on assessment by the MAME developers themselves. We created a binary variable *MAJOR_MFG* with a value of 1 if a game was made by any of the 12 largest game manufacturers of the arcade era (see table 2). The variable *GENRE_SHOOT* is a dummy variable with a value of 1 if a game is in the “shooter” genre (arbitrarily chosen), to assess whether genre has any effect on timing of MAME support. Finally, the variable *INPUT_NONSTD* is a dummy variable with a value of 1 if the game had non-standard input, with the reference category being the standard 8-way joystick.

Descriptive statistics for the sample of 3783 arcade machines					
Variable name:	Range:	Min:	Max:	Mean:	Std. Deviation:
GENRE_SHOOT	2499.99	.00263	2500	9.99	52.013
GAME_AGE	40	5	45	32.32	7.227
MAME_AGE	26	1	27	19.58	6.297
YEARS_DIFF	44	0	44	12.42	6.460
TPM_RECOVERY	1	0	1	.19	.388
PROBLEM_GRAPH	1	0	1	.23	.423
PROBLEM_SOUND	1	0	1	.33	.469
PROBLEM_PROTCT	1	0	1	.02	.150
MAJOR_MFG	1	0	1	.54	.498
GENRE_SHOOT	1	0	1	.30	.457

Table 3: Descriptive statistics for the sample of 3783 arcade machines



TRY 2P COOP

1st	CAP	237700	1 MIN
2nd	CAP	182500	0 MIN
3rd	CAP	70700	0 MIN
4th	NIN	50000	10 MIN
5th	ADI	300	1 MIN

TRY 2P COOP

1st	VDI	800	0 MIN
2nd	NIN	20000	10 MIN
3rd	CUB	30300	0 MIN
4th	CUB	183200	0 MIN
5th	CUB	303200	0 MIN

5. RESULTS

We subjected the data to statistical analysis to better understand the use of TPMs in arcade video game hardware, and to assess the impact of TPMs on the preservation efforts of MAME volunteers. Table 4 summarises the distribution of TPMs on arcade hardware in the full MAME 0.258 supported devices list, as well as the sub-sample of all arcade games released from 1979–2023.

We find that TPMs were extensively applied by manufacturers of arcade games, with 1,072 games or 29% of the sample having some form of protection. It is likely that the true number of TPM-protected games is higher, because our keyword-based collection method only captures titles where the presence of TPM protection was noted by the MAME developers. The most extensively specified form of protection was encryption of game data, found on 370 devices or 10% of the arcade sample. A similar number of boards utilised protected Memory Controller Units (MCUs). It should be noted that there is overlap in the reported numbers for specific protection techniques – some games were protected using multiple techniques or were detected in multiple ways because MAME developers used different language to describe protections. We probably under-sampled the total number of devices protected by TPMs because those with trivially circumvented protections may not have warranted a written entry from developers.

Summary statistics on the presence of TPMs in full MAME, device list and arcade only sample				
	All Devices n=14377	%	Arcade Only n=3783	%
TPM present	1430	(9.95)	1072	(28.89)
Type: DS5002FP	13	(0.09)	11	(0.30)
Type: FD1094	55	(0.38)	51	(1.37)
Type: Suicide	10	(0.07)	8	(0.22)
Type: Checksum	103	(0.72)	69	(1.86)
Type: Encryption	541	(3.76)	370	(9.97)
Type: Security chip	48	(0.33)	42	(1.13)
Type: MCU	486	(3.38)	335	(9.03)
Type: Unspecified prot.	871	(6.06)	701	(18.89)
TPM recovery used	904	6.29	712	18.82
Type: Simulation	304	(2.11)	262	(6.93)
Type: Decapping	192	(1.34)	155	(4.10)
Type: Dump protection	714	(4.97)	565	(14.94)
Type: Hack protection	179	(1.25)	147	(3.89)
Problem: Graphics	1862	(12.95)	901	(24.28)
Problem: Sound	6193	(43.08)	1201	(32.36)

Table 4: Summary statistics on the presence of TPMs in full MAME, device list and arcade only sample

Because we are interested in measuring the resource costs associated with circumvention of TPMs, we attempted to isolate games where circumvention required active effort by MAME contributors. Some TPMs were trivially circumvented and therefore did not impose additional costs on preservation. However, active circumvention of TPMs was required in 712 cases (19% of games). Various circumvention techniques were reported by MAME developers. Simulation of the protection measure was required to successfully emulate games in 262 cases (7% of sample). This technique involves reproducing the behaviour of an underlying TPM in order not to trigger any copy protection restrictions by the main game code as it runs. In 565 cases (15% of sample), developers acquired the game's physical hardware and were able to dump the contents of any protected code, circumventing the hardware locks. In 155 cases (4% of sample), microprocessors needed to be additionally decapped (physically opened) and their encrypted contents extracted before the TPM could be fully understood and reproduced in emulation. A typical means of decapping involves physically opening a protection module, applying nail polish or another UV-blocking substance to the IC containing the protected code, and then erasing the encrypted lockbit with UV light to enable access to the protected contents. As described in section 3 above, some methods of decapping were more elaborate, depending on the sophistication of the TPM. Again, it should be noted that there is overlap in the circumvention techniques reported. Some games might have initially been emulated by hacking the functionality of a TPM module, and later decapped to enable full preservation and emulation of the board. In such cases, the date of addition to MAME reflects the earliest successful circumvention of the TPM, which means that we consistently report the lowest measurement of years between initial release and first emulation in MAME, rather than time to "perfect" emulation.

Next, we performed a series of ordinary least squares (OLS) regressions to understand the relationship between our study variables and the span of years between initial game release and successful preservation in MAME (measured by *YEARS_DIFF*). Table 5 shows results of the regressions with controls added progressively in each of the model specifications. We report unstandardised coefficients to aid interpretation. We find that circumvention of TPMs is significantly and positively associated with *YEARS_DIFF*, indicating a delay in successful emulation caused by the need to circumvent TPMs. With all controls included in the model, the coefficient for total delay caused by TPMs is .882 years (10.6 months) per game requiring circumvention. Multiplied by the total number of games in our sample requiring circumvention, we obtain a total social cost from TPMs for the MAME project of $0.882 * 712$ games, or 627.9 years of total cost/delay. This figure represents the additional effort required from MAME developers to circumvent TPMs, as well as the missing use in terms of research and cultural benefit lost from the delay to successful preservation caused by TPMs.

The direction and significance of the effect is robust across the specifications, while model fit also gradually increases with addition of more controls. The effect is also robust when data is sliced by different time scales, as discussed further below. The variable *ROMSIZE* is negatively associated with *YEARS_DIFF*, reflecting the emergence of larger and more complex games in recent history. We control for the evolution of PC hardware capability using the variable *LOG_GFLOPS*. The control variables *PROBLEM_GRAPH* and *PROBLEM_SOUND* are also significantly associated with *YEARS_DIFF*, indicating that these issues are associated with the speed of emulation support. Problems with copyright protection were significantly associated with an increase in *YEARS_DIFF*, reinforcing the effect of TPMs on increased development time before successful emulation. Games by major arcade manufacturers were negatively associated with *YEARS_DIFF*, indicating that their games are more recent or more swiftly added to MAME due to their popularity.

TPM circumvention and delay from initial game release to preservation				
Dependent variable:	(1) YEARS_DIFF	(2) YEARS_DIFF	(3) YEARS_DIFF	(4) YEARS_DIFF
TPM_RECOVERY	.989*** (.157)	.991*** (.156)	.912*** (.220)	.882*** (.157)
LOG_ROMSIZE	-1.707*** (.031)	-1.785*** (.035)	-1.741*** (.037)	-1.724*** (.037)
LOG_GFLOPS	1.725*** (.082)	1.742*** (.081)	1.736*** (.080)	1.781*** (.080)
PROBLEMS_GRAPH		1.291*** (.211)	1.293*** (.214)	1.248*** (.211)
PROBLEMS_SOUND		-.659*** (.180)	-.680*** (.180)	-.580*** (.177)
PROBLEMS_PROTCT			1.365* (.549)	1.365* (.541)
IS_MAJOR_MFG			-.544*** (.143)	-.463** (.142)
INPUT_NSTD				-.647** (.216)
GENRE_DUMMY				.918*** (.151)
_constant	34.900*** (.434)	35.548*** (.470)	35.554*** (.493)	35.078*** (.502)
N	3783	3783	3783	3783
Adj.- R²	.571	.576	.575	.580

Table 5: TPM circumvention and delay from initial game release to preservation
Unstandardised coefficients reported / Standard errors in parentheses / * p < 0.05, ** p < 0.01, *** p < 0.001

We performed a series of robustness checks to ensure that the assumptions of the linear regression model were met. Residuals were normally distributed, upholding the linear assumption of the OLS regression approach. We performed multicollinearity diagnostics and did not find strong collinearity between independent variables. All VIF scores were under 1.5 and condition index scores calculated from eigenvalues were all below 15, indicating no significant multicollinearity. We report heteroskedasticity-robust standard errors in all models.¹⁵

¹⁵. Heteroskedasticity refers to a feature of certain relationships in which the variance of the error term is non-constant. In this case, variance in Years_Diff increases over the MAME development history as a range of newer and older games became available for emulation. We address this by reporting robust standard errors which reflect the additional variance to ensure that standard errors and p values are accurately represented.

Next, we investigated whether delays related to the circumvention of TPMs varied at different times across the history of MAME development. We do this to test 2 propositions: (1) That the delay from TPMs increased over time due to the difficulty of particularly hard cases remaining to be emulated; and (2) that the effect of TPMs on successful preservation might differ between historical and contemporaneous preservation (historical preservation referring to games that were released before the MAME project began compared to games released while the emulator was in active development). To test proposition 1, we split the sample into two equal time periods based on when a game was first supported in MAME. During the first period of MAME development from 1997–2009, the project successfully added support for 3,103 arcade games. During the later period of development from 2010–2023, the project added support for 680 games. The rate of addition of new titles has slowed as fewer arcade games remain to be successfully preserved. MAME developers have turned their attention to other types of hardware, adding support for handheld devices, personal computers and other non-arcade machines in recent years.

Table 6 shows the results of OLS regressions on the dependent variable *YEARS_DIFF*, when the sample is split into the respective development periods (early: 1997–2009; and late: 2010–2023). We see in columns 1 and 2 that the effect of TPM circumvention delay is present and statistically significant during early MAME development, with an average delay of .992 years or 11.9 months of delay added when circumvention was required. In the second half of MAME development (from 2010–2023) the observed delay from circumvention is higher (1.209 years or 14.5 months per game). However the variance, indicated by the standard error, is also higher and the effect from TPMs is not as significant overall compared to early MAME development. We interpret this to indicate that certain games with complex TPMs are taking longer to successfully preserve in more recent MAME activity, while others are more rapidly circumvented, perhaps due to the accumulated experience of MAME contributors or the availability of new methods to facilitate circumvention.

Delays from TPM circumvention during early and late MAME development				
Dependent variable:	(1) YEARS_DIFF	(2) YEARS_DIFF	(3) YEARS_DIFF	(4) YEARS_DIFF
Sub-sample:	1997–2009	1997–2009	2010–2023	2010–2023
TPM_RECOVERY	1.050*** (.133)	.992*** (.133)	1.494** (.552)	1.209* (.584)
LOG_ROMSIZE	-1.839*** (.031)	-1.832*** (.031)	-1.799*** (.104)	-1.816*** (.103)
LOG_GFLOPS	.457*** (.046)	.510*** (.046)	1.484*** (.087)	1.700*** (.069)
PROBLEMS_GRAPH	.205 (.193)	.116 (.193)	2.010*** (.510)	1.451** (.525)
PROBLEMS_SOUND	-.522** (.160)	-.577*** (.157)	-5.521*** (.452)	-4.146*** (.508)
PROBLEMS_PROTCT		1.240** (.402)		-1.191 (.763)
IS_MAJOR_MFG		.526*** (.117)		-1.430*** (.447)
INPUT_NSTD		.906*** (.118)		-2.564*** (.560)
GENRE_DUMMY		.890*** (.122)		1.585** (.535)
_constant	39.814*** (.715)		45.157*** (1.419)	46.032*** (1.412)
N	3103	3103	680	680
Adj.- R²	.632	.642	.621	.647

Table 6: Delays from TPM circumvention during early and late MAME development

Unstandardised coefficients reported / Standard errors in parentheses / * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

We visualise the variance in *YEARS_DIFF* for games requiring circumvention in Figure 3. As can be seen, the pattern corresponds to the results obtained in our regression analysis: *TPM_RECOVERY* is significantly associated with an increase in *YEARS_DIFF* across the lifetime of MAME development, but there is increasingly more variance in *YEARS_DIFF* as we approach the present year. We interpret this to reflect the reality of carrying out digital preservation on a limited population of historical materials. In the first period of MAME development, the population of un-emulated games was greater, and developers could choose from among a long list of candidate games going back to the 1970s. As time went on, there remained fewer arcade games to emulate, with some protected games requiring extensive effort to circumvent. The effect from TPMs might be more pronounced in the later years of the project because the “low hanging fruit” was successfully preserved and only lesser-known and more difficult hardware remained to be emulated. To test proposition 2, we also examined whether modern games (released after 1997) exhibited any difference from historical games in terms of delay. We found that the delay from TPMs was still present and significant for both historical and contemporaneously preserved games. Modern game releases protected by TPMs were associated with a similar amount of delay compared to older titles (approximately 11 months), suggesting that the complexity of TPMs has kept pace with the ability of determined individuals to circumvent them.

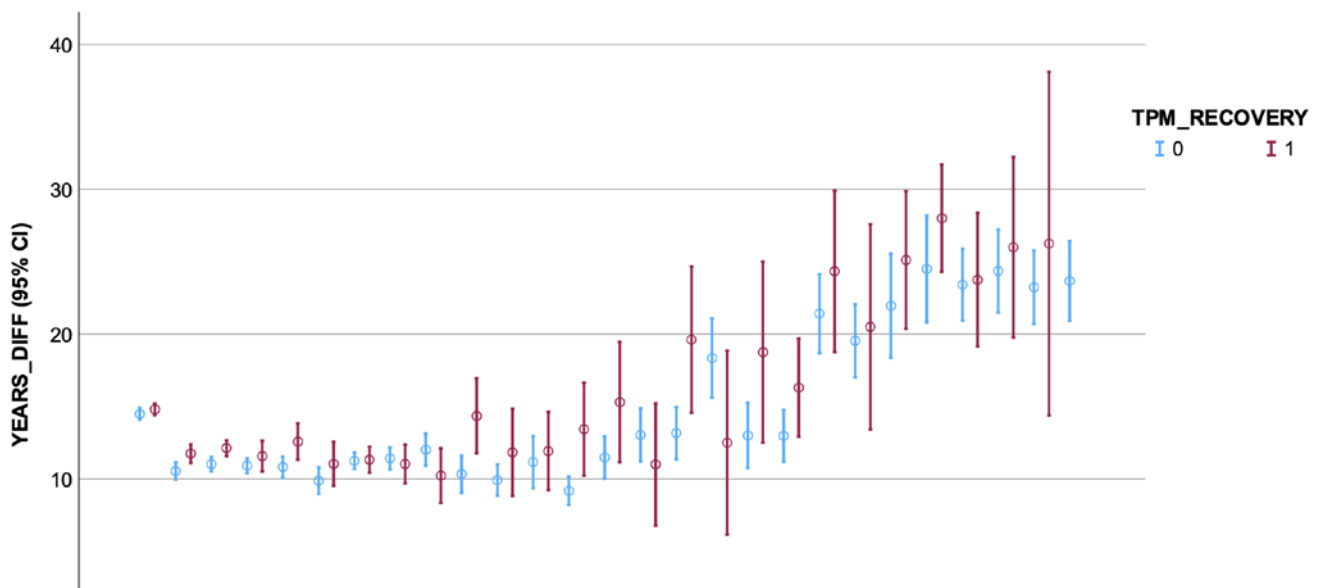


Figure 3: Variance in *YEARS_DIFF* by presence of TPM circumvention during MAME development



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6. RECOMMENDATIONS

The global video games industry generated US\$281bn in 2023, more than double the estimated global market for film (Fortune, 2023). The economic and cultural significance of interactive games is expected to continue to grow, while the legal and regulatory frameworks supporting preservation of these materials is widely argued to be inadequate (Lee, 2018; Lemley, 2021; Farrand, 2022). Based on our analysis above, we have established that TPMs caused significant practical impediments to preservation of video games.

There are also significant legal impediments for researchers and preservation institutions as we discuss below. The presence of TPMs on arcade hardware contributed an average of .882 years per game and an aggregate 628 years of additional time delay to the overall MAME project, controlling for other factors such as game complexity, manufacturer, and compute resources available to preservationists. These delays represent significant losses to research, education, follow-on innovation and cultural heritage. The persistent challenges raised by TPMs present major technical and legal obstacles to beneficial uses such as research and preservation.

There are two distinct social costs represented by the aggregate time delay that we measured, relevant for a range of born-digital materials, not only video games: (1) a portion of the time delay represents additional human labour required to locate a surviving example of material, identify any TPMs present in the device, and figure out a method of circumvention to enable preservation. It should be emphasised that this option is unavailable to those unwilling to infringe copyright law in jurisdictions where circumvention is prohibited. (2) A second social cost is represented by the delay in availability to follow-on uses such as research or innovation caused by the presence of TPMs. For example, consider two similar games released in the year 1985, one with TPMs and one without. The game released with TPM protection likely took longer for MAME developers to access and make available, as demonstrated in our study. That time delay represents a period in which the game protected by TPMs was unavailable to research, preservation and other follow-on uses compared to the game without TPMs. There are an unknown number of games still awaiting preservation that are locked behind TPMs, and inevitably some of those will be lost forever. Currently, 162 devices in MAME are only partially functional due to enduring problems with protection, causing glitches and crashes upon playback. This means that we are likely to underestimate the social cost from TPMs in this report, since ongoing efforts spanning decades may ultimately fail to perfectly emulate some TPM-protected devices.

While our results represent the first quantitative evidence on the impact of TPMs on preservation, the problem has already been identified as an urgent concern by practitioners (Lowood et al., 2009; Yee, 2021). Based on our findings and those echoed by practitioners in cultural heritage, research and industry, we have two main recommendations: (1) to clarify that cultural heritage institutions may circumvent TPMs for preservation purposes without having to navigate the administratively cumbersome processes currently required by European copyright law;¹⁶ and (2) that measures be taken to offset the social costs of TPMs for preservation and research by requiring publishers to take necessary steps to enable access and reduce negative spillovers.

¹⁶ See Report 1: Rosborough, A. (2024) Technological Protection Measures and the Law: Impacts on Research, Education and Preservation. Fig. 11.

Recommendation 1:

Specific exemptions to anti-circumvention should be crafted to enable European research institutions, libraries and archives to catch up to the current state of the art in emulation and benefit from the achievements of volunteer initiatives like MAME. As shown in a survey of research and preservation activities,¹⁷ institutions are highly risk-averse in dealing with TPMs, and may avoid using emulation tools such as MAME due to concerns about copyright. However, in many cases, emulation is the only practical method of retrieving and making available legacy digital materials on obsolete formats.

TPMs present the main legal impediment to use of emulation by preservation institutions and researchers. Under existing EU copyright law, both emulation of hardware and backing up of lawfully-acquired copies for preservation likely do not infringe copyright. Emulation of hardware usually involves reverse engineering and examination of embedded BIOS software. This activity probably falls under Software Directive Article 6, being done for the purposes of achieving interoperability of an independently created computer program with other programs (van der Hoeven et al., 2010). Similarly, as pointed out by Farrand (2012) video game emulators appear to meet the requirement of enabling software-to-software interoperability under the Software Directive. Further, Article 6 CDSM Directive provides for a mandatory exception “to permit cultural heritage institutions to reproduce works and other subject matter permanently in their collections for preservation purposes, for example to address technological obsolescence [...] by the appropriate preservation tool, means or technology, in any format or medium, in the required number, at any point in the life of a work or other subject matter and to the extent required for preservation purposes”. The problem for preservation is that the CDSM Directive is not explicit about whether this exception permits circumvention of TPMs required to accomplish the task of reverse engineering and preservation.

While Article 6(4) InfoSoc Directive instructs Member States to ensure the means are available for prospective beneficiaries to request access to TPM-protected materials from rightsholders, this may be impossible where rightsholders cannot be located, or where third-party manufacturers of TPMs are no longer in business (giving rise to orphan TPMs). In the case of video game preservation, many of the manufacturers of devices emulated in MAME are no longer in business or are no longer interested in exercising their rights. Administrative processes to request removal of TPMs are cumbersome, and costs are likely to multiply to unacceptable levels for mass digitisation initiatives.¹⁸ Unnecessarily hindering CHIs should be weighed carefully in a context where MAME already exists, and volunteer repositories of nearly all historical video games from 1960 to the present are already accessible online.

Building upon earlier recommendations by Hoeven et al. (2010) we suggest that EU policymakers remove the requirement for cultural heritage institutions to seek access from rightsholders for circumvention of TPMs when engaging in preservation as defined under Article 6 CDSM Directive. While few libraries or museums might seek to develop their own bespoke emulators or circumvention tools (since these already widely exist), clarification of the ability to circumvent TPMs for preservation would embolden institutions to make more innovative use of these technologies to preserve digital materials (residing on circuit boards, magnetic disks and other obsolescing formats) that they permanently hold in their collections.

¹⁷. See Report 2: Erickson, K. and Stobo, V. (2024). Survey on Technological Protection Measures: Impacts for Researchers, Libraries and Archives.

¹⁸. See Erickson K., & Stobo, V. (2024). Survey on Technological Protection Measures: Impacts for researchers, libraries and archives. KR 21 Project Report 2.

Under the DMCA triennial rulemaking in the USA, eligible libraries, archives, and museums are permitted to circumvent TPMs for preservation purposes on lawfully acquired computer programs (including video games) when they are no longer reasonably available in the marketplace. The wording of this exception would cover most of the legacy game devices preserved in MAME (although MAME itself is not eligible for the exemption as a volunteer initiative rather than an institution). Under the recent DMCA exemption rulemaking of 2021–2024, institutions in the USA may additionally make single preserved copies available off-premises to a single user at a time. Libraries, archives and researchers in Europe need similar legal clarity and the ability to use existing software tools such as MAME to undertake the socially beneficial legal activities that are already covered by exceptions to copyright.

Recommendation 2:

We recommend that future policy making more fully account for social costs imposed by TPMs when assessing their impact. Manufacturers and publishers have failed to provide strong economic evidence that TPMs are necessary to protect and successfully commercialise their intellectual property. In fact, robust economic research indicates the opposite: there are significant commercial benefits to relaxing or removing TPMs to permit interoperability, adding value for consumers and society (see for example market studies by Dorr et al., 2009; Erickson et al., 2017; Volckmann, 2024). Rothchild (2006) likens the negative externalities resultant from TPMs to those caused by pollution, suggesting policy intervention to ensure that producers bear the costs for long-term effects of TPMs. Policy makers might consider setting time limits on TPMs to ensure that works are accessible to all once locks become obsolete. These limits should reflect real-world market exploitation of works and be designed to facilitate research and preservation uses. Manufacturers and publishers could be required to design work-arounds when using TPMs to facilitate enjoyment of exceptions, accessibility and interoperability. One of the main findings of this study is that legacy TPMs can still cause significant impediments. When knowledge about implementation of a specific TPM is lost to the passage of time, contemporary researchers must invest significant labour to reverse engineer them.

National lawmakers and libraries can exert pressure to ensure that contemporary TPMs applied to new works are removable for purposes of future preservation. As discussed by Hoeven et al. (2010), this is already the case in certain jurisdictions. For example, legal deposit of multimedia works at the Bibliothèque Nationale de France must be accompanied by TPM access codes.¹⁹ The Code of the German National Library requires that legal deposits must not be blocked by TPMs. If the TPM cannot be removed, the access codes should be provided (while this applies to multimedia works, it does not include video games which are excluded from the legal deposit requirement).²⁰

There have been calls for greater collaboration between industry and the preservation sector (Lowood et al., 2009; Bachell & Barr, 2014; Stuckey & Swalwell, 2014). Volunteer collective projects like MAME should also be part of this process. The games industry benefits from the existence of MAME and other formal preservation efforts: preservation legitimises video games as culturally significant; museums, archives and libraries serve as repositories of innovation and inspiration for future production, and they enable research and learning about software and hardware development.

¹⁹. Source: <https://maint.loc.gov/law/help/digital-legal-deposit/france.php>

²⁰. Source: <https://maint.loc.gov/law/help/digital-legal-deposit/germany.php>

Companies have used MAME and other volunteer open-source emulator projects as the basis for new products, such as the Evercade handheld console manufactured by UK company Blaze Entertainment, which offers consumers the ability to play licensed emulated ROMs re-released by golden age arcade publishers. The Capcom Home Arcade controller manufactured by Koch Media used emulation based on open-source software to play licensed arcade games packaged with the device. In both cases, the availability of open-source emulation software and preserved ROM files enabled manufacturers to commercially exploit legacy games from their catalogue. If emulators did not exist, the prohibitive cost of bespoke software development might have prevented release of these niche products. Productive collaborations with industry might involve establishment of open data pools to collect and share information about historical games, encouragement of emulation as a preservation strategy for traditional institutions, and clarification of acceptable norms of practice involving copyright materials to enable more preservation and use by CHIs. These should be as open and encouraging as possible, given the progress already made by informal communities like MAME, which are widely available to users online.

To date, after successful emulation of 14,377 legacy hardware devices and circumvention of over 900 obsolete TPMs, there have been no legal challenges made against the MAME project by copyright owners. The games industry has tolerated emulation while it has also benefited from it as a resource. MAME should not only be tolerated, but its methods widely celebrated and encouraged as an effective preservation approach. The clarifications that we recommend to EU copyright law to enable circumvention for preservation would level the playing field for institutions, enabling them to contribute their resources and expertise to the benefit of European culture and innovation.



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7. LIMITATIONS AND DIRECTIONS FOR FUTURE RESEARCH

We have outlined a methodology for studying the social costs (represented by human labour and time delays) imposed by TPMs on downstream uses such as research and preservation. Our aim is that the approach outlined here can be of wider use to researchers of born-digital preservation (including neighbouring media like digital art, multimedia, electronic documents and software).

Other media may not benefit from enthusiastic volunteer communities such as MAME to assist with circumvention, so effects from TPMs may be even more profound for institutions lacking technical expertise to preserve materials. A more significant factor is unwillingness of CHIs to violate copyright law. Volunteer communities, especially international ones, may feel less encumbered by the inflexibility of EU copyright law than European cultural institutions. In many ways, the MAME study is a “best case” scenario for preservation, due to the popularity of gaming, the dual-purpose of MAME as archive and entertainment, and the relatively small number of unique hardware devices requiring preservation (14,377 devices in MAME, compared to potentially billions of electronic records stored on legacy formats).

While we hope this study serves as a template and instructive base of evidence, there are limitations related to generalisability. This study focused on a single preservation initiative and a single media type. The social costs represented by TPMs are probably more widespread and impact different institutions and communities in various ways. Emulation is not always a suitable approach: for example, preservation of digital artwork sometimes involves concerns about authenticity and experiential aspects that are harder to reproduce with emulation (see Stuckey & Swalwell, 2014; Ippolito, 2016).

Future research could generate empirical measurement of costs for the preservation of other media such as film, electronic documents, images, software and IoT devices, where TPMs are frequently present. However, doing so may be challenging due to limited visibility of resource costs compared to MAME, where careful documentation facilitated our quantitative data collection. Surveys, guided interviews, documentary and case study methods may be applicable depending on the institutional context. Since many born-digital materials lack the benefit of volunteer preservationist communities, this makes the understanding the task of digital preservation even more urgent and challenging.

More research can be done to assess the impact of copyright TPMs on video game preservation, including challenges introduced by software-as-a-service (SaaS) distribution models, which we did not address in this report.²¹ Newer online and mobile games often require users to be logged in to a server to play, meaning that preservation of these experiences would require reproduction of server software to preserve functionality. This may involve circumvention of TPMs used for server authentication or encryption, frequently used in multiplayer and mobile games (Lee, 2018).

²¹ In 2024 a European Citizens Initiative petition was launched, seeking “to prevent the remote disabling of videogames by publishers, before providing reasonable means to continue functioning of said videogames without the involvement from the side of the publisher”. The initiative had raised over 360,000 signatures as of October 2024. Source: https://citizens-initiative.europa.eu/initiatives/details/2024/000007_en

To further isolate the impact of copyright TPMs on preservation of various devices, future analyses could add controls for standardised hardware components (e.g. popular chipsets) to enhance the analysis provided here. Our study controlled for this using major manufacturers as a proxy variable, since game manufacturers regularly made use of standardised hardware. We also controlled for emulation difficulty using data about graphics and sound issues encountered during preservation, and for overall complexity by using game ROM memory size as a proxy. We tested (but did not report) the relationship between manufacturer size and TPM use, expecting to observe differences. Surprisingly, we did not find significant dissimilarities or idiosyncrasies related to TPM use by smaller manufacturers, suggesting the dominance of established hardware formats in the arcade market. One notable exception was Gaelco, whose use of the DALLAS protection significantly hindered preservation efforts. Overall, the concept of social costs provides a framework for future study of the effects of copyright protection for preservation and access to knowledge across the full range of copyright industries, of which video games is only one example.



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8. CONCLUSIONS

We have analysed the effect of TPMs on preservation of digital cultural heritage, using the MAME emulation project as a study site. We found that the presence of TPMs on legacy arcade video game hardware significantly delayed successful preservation, adding an average of .882 years or 10.6 months to MAME development time for each TPM-protected game requiring circumvention. We probably under-estimate the delay in fully preserving games, since initial support might still be imperfect, relying on hacks and other techniques for partial preservation. Our report also excludes work undertaken on TPM-protected games that have not been successfully emulated to date.

To avoid false positives and to present the most conservative estimate possible, we applied a strict criterion to detect games where active circumvention of TPMs was required (those where developers noted both the presence of protection and a specific method of circumvention in the development notes). The effect of TPMs on the overall MAME emulation project is more widespread than we report here, also impacting upon non-arcade hardware included among the more than 14,300 devices currently emulated by the MAME project.

Our study presents the first economic estimate of the social cost of TPMs in digital preservation. The MAME preservation effort is unique compared to traditional archival institutions because MAME volunteers have not been inhibited by copyright law, while memory institutions are far more constrained. If MAME and similar emulation projects did not exist, it is likely that the situation regarding preservation of 20th century video games would be much more dire. Deterred by both legal uncertainty and technical challenge, it is likely that traditional CHIs would have avoided such a project.

While costly in terms of human effort, emulation initiatives like MAME generate positive externalities: some traditional CHIs have used emulation to restore and exhibit digital artworks;²² firms have used MAME's open-source code to enable new products based on legacy video games, communities of enthusiasts have sprung up around retro games generating new economic activity such as tournaments and complementary volunteer initiatives.²³ These benefits could be even more numerous and could have arrived sooner if TPMs had not delayed preservation efforts.

The main things that policy can do to ameliorate the problem are: (1) to clarify that cultural heritage institutions may circumvent TPMs for purposes of preservation without having to navigate the cumbersome administrative processes currently imposed by European copyright law;²⁴ and (2) to weigh the future costs of TPMs when balancing strong protections granted to rightsholders, for example by requiring that rightsholders maintain and offer robust means for European CHIs and researchers to access TPM-protected works into the future.

As digital preservation expands to address more forms of digital media, undoubtedly new and challenging TPM methods will be encountered. Unlocking obsolete TPMs is within the ability of skilled programmers, memory institutions and dedicated communities, but not without significant cost.

²² In addition to the Guggenheim example discussed by Jones (2004), Stuart Brown used emulation to facilitate digital recreation of the work *Four-Byte Burger* (1985) by Jack Haegar, a work that existed only in analogue form after the original file was lost. Source: <https://www.youtube.com/watch?v=i4EFkspO5p4>

²³ For example, the MAME Action Replay Page contains over 15,000 historical recordings of individual gameplay from the early 2000s. Source: <http://replay.marpirc.net/test/index.htm>

²⁴ See Report 1 by Rosborough A. (2024).

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Appendix 1: Glossary of terms

Glossary of terms	
Term:	Definition:
Checksum	▶ A method of securing software that involves checking the size of a file (for example sounds on a ROM chip) to ensure that it hasn't been altered.
Copyright	▶ In the context of this study, any form of technical copyright protection noted by the developers.
Decapping	▶ A method of circumventing TPMs in which microprocessors have their coverings removed (decapped) to expose the circuits and embedded firmware underneath.
DS5002FP	▶ A specific security module sometimes referred to at the 'DALLAS' semiconductor, which used a battery and encryption to prevent running a game's code on any other hardware.
Dump	▶ The process of backing up game files contained on ROM chips, usually accomplished with an EPROM programmer device.
Encryption	▶ In the context of this study, any form of encryption used by game manufacturers intended to prevent access or copying of game code.
FD1089/94	▶ A specific series of security chips manufactured by Hitachi and used in many Sega arcade games. These are 68000 processor cores with a battery and encryption keys used to decrypt the game at run-time.
Hack	▶ In the context of this study, a TPM recovery method involving circumventing or bypassing a security module, rather than perfectly emulating it.
MCU	▶ Memory Controller Units (MCUs) are digital circuits that manage the flow of data going to and from a computer's main memory. They can operate as TPMs, for example by providing memory scrambling.
Protection	▶ In the context of this study, any generic copyright security protection noted by the MAME development team.
ROM	▶ Read-only memory (ROM) refers to hard-wired memory, such as a mask ROM integrated circuit (IC), that cannot be electronically changed after manufacture. Many arcade and home console games distributed on cartridges used ROMs for storage of game contents.
Security	▶ In the context of this study, any generic copyright security protection noted by the MAME development team.
Simulation	▶ In the context of emulation, one approach to circumvention involves simulating the behaviour of a TPM module without necessarily perfect understanding its internal code or architecture.
Slapstic	▶ Colloquial name among arcade preservationists for a series of integrated circuits used by Atari to secure early arcade games.
"Suicide" hardware	▶ Colloquial term within the arcade preservation community for hardware TPMs that wipe board contents if tampered with, usually achieved via an inaccessible battery.